

MSAI 699 Capstone Project Capstone

Week 3: Model Selection and Baseline

NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence Platform for Personalized Cognitive Health Education Using Machine Learning and Retrieval-Augmented Generation

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Introduction

This report shows the baseline machine learning model built for the risk assessment tool in the *NeuroGuide AI*. In line with the project proposal and the literature review conducted in Week 1 & Week 2, the model was built using the Alzheimer's Prediction Dataset (Global) (Panday, 2025) to show that there is an initial risk classifier that can be developed even before any hyperparameter tuning, imbalanced classes correction and integration with the Retrieval-Augmented Generation & multi-agent features of the platform. The full implementation process is included in the attached Jupyter notebook (week3_risk_model.ipynb).

Model Architecture & Justification

The chosen baseline is a Random Forest classifier (Breiman, 2001). This choice was motivated by the following benefits mentioned in Week 2 literature review: robustness to heterogeneous and noisy tabular data, protection against overfitting with an increase in the number of trees used, built-in variable importance estimates, and good SHAP (Lundberg & Lee, 2017) compatibility.

Starting from the raw dataset with 74,283 observations and 25 features without any missing values, twelve features have been selected based on the criteria of being modifiable or non-modifiable dementia risk factors provided by the Lancet Commission (Livingston et al., 2024): age, years of education, presence of genetic APOE- ϵ 4 risk factor, hypertension, obesity (calculated using BMI), current smoking status, depression, physical inactivity, diabetes, lack of social interaction, excessive alcohol consumption, and hypercholesterolemia.

To better understand the dataset before model development, exploratory data analysis (EDA) was conducted to examine the distribution of the target variable and key predictor variables. Figures 1–3 summarize these findings.

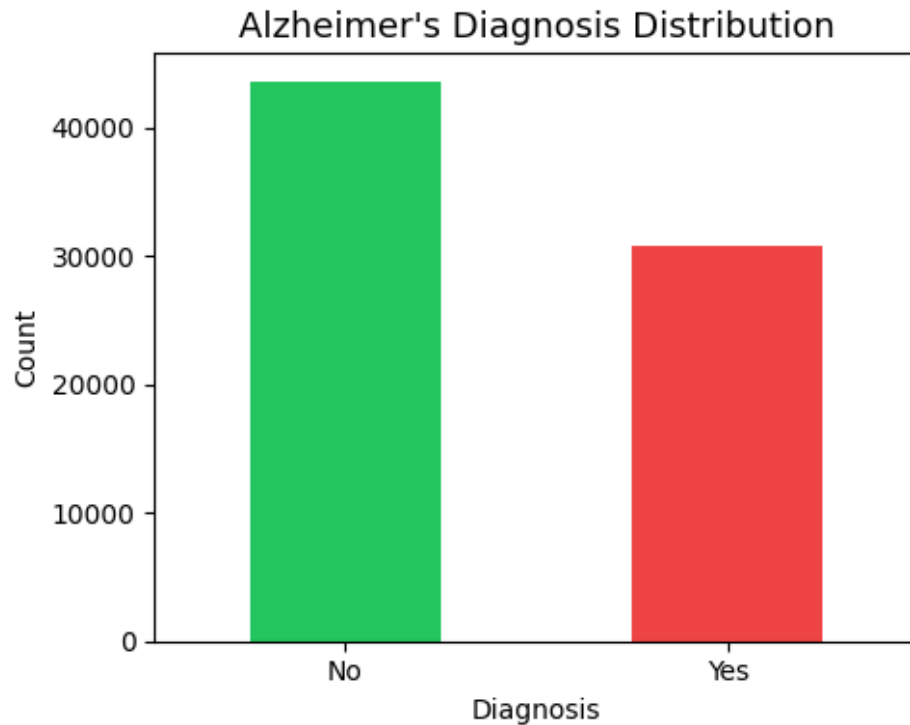


Figure 1. *Distribution of Alzheimer's Diagnosis in the Dataset*

Figure 1 demonstrates that the dataset is relatively balanced, with slightly more participants without an Alzheimer's diagnosis than those diagnosed.

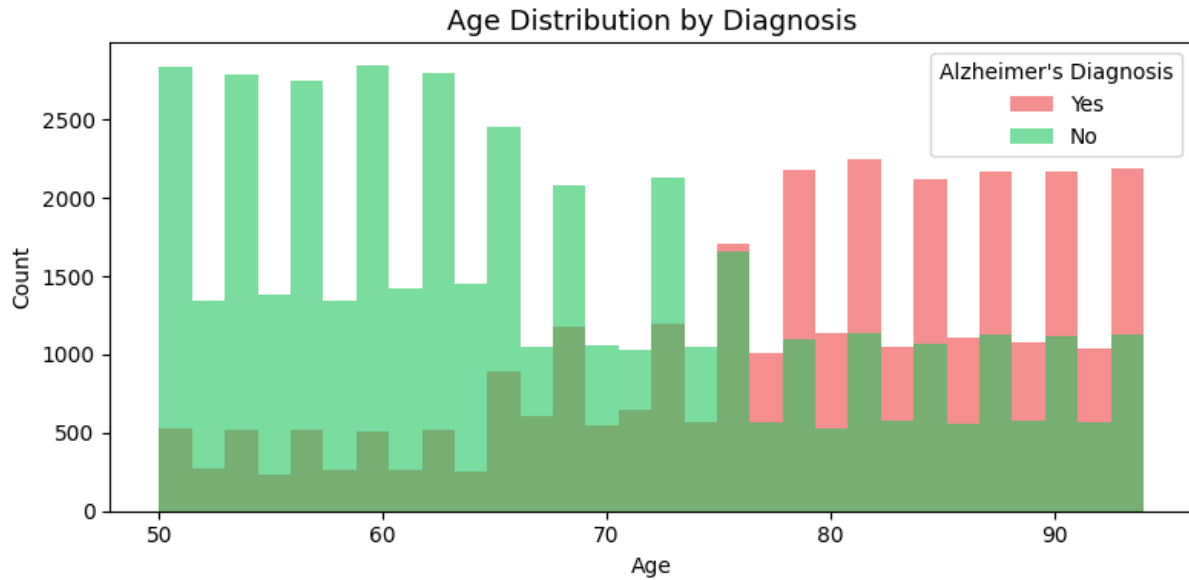


Figure 2. *Age Distribution by Alzheimer's Diagnosis*

Figure 2 shows that diagnosed individuals generally appear at older ages, supporting age as one of the strongest known Alzheimer's risk factors.

Cognitive test score also showed meaningful separation between diagnosed and non-diagnosed participants. As shown in **Figure 3**, individuals with Alzheimer's diagnosis generally exhibited lower cognitive assessment scores, providing additional justification for incorporating cognitive performance into the educational risk categorization used by NeuroGuide AI.

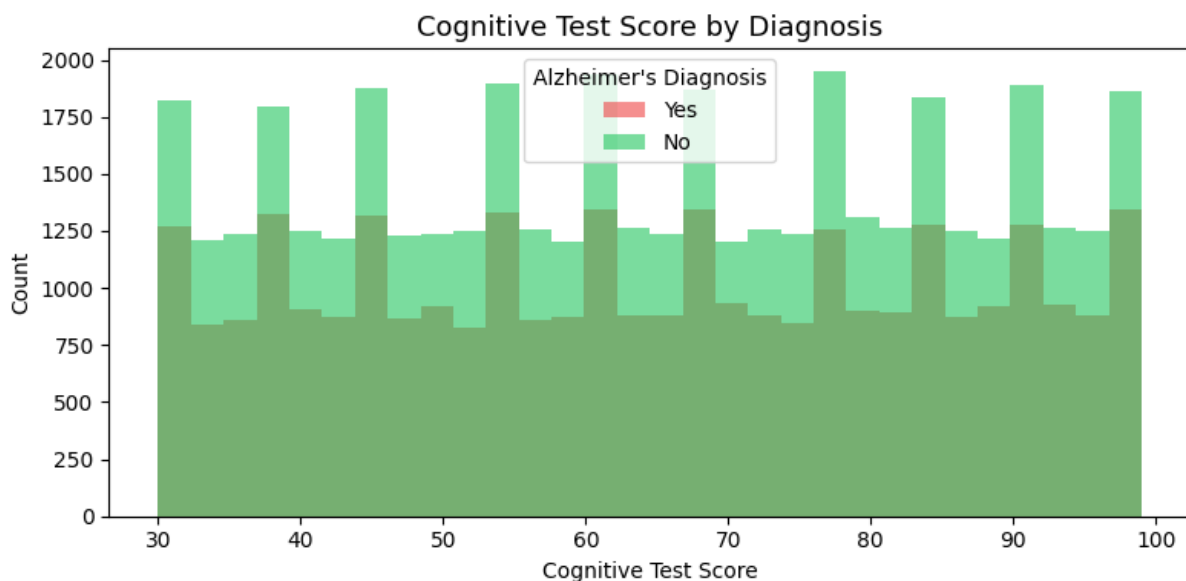


Figure 3. *Cognitive Test Score Distribution by Alzheimer's Diagnosis*

Figure 3 illustrates that cognitive test scores differ across diagnostic groups, further supporting the inclusion of cognitive performance when constructing the educational risk categories used by the baseline model.

Due to the fact that the target variable is a binary (Yes/No) variable for the presence of Alzheimer's, it has been transformed into a three-category (Low, Moderate, High) risk level classification system using the diagnosis result together with the cognitive test score from the dataset. Such transformation makes it easier to interpret the results because NeuroGuide AI is an educational and not diagnostic product. The class distribution is relatively well-balanced between Low (42.1%), High (41.3%) and Moderate (16.5%). There were 200 decision trees used, the maximum depth is 8 in order to prevent overfitting, and balanced class weights have been used due to lower numbers in the third class.

Initial Accuracy & Baseline Performance

On the held-out test set (14,857 samples), the baseline model produced the following metrics:

Metric	Value
Accuracy	0.516
Precision (weighted)	0.544
Recall (weighted)	0.516
F1 Score (weighted)	0.519
ROC-AUC (one-vs-rest)	0.718
5-Fold CV F1 (weighted)	0.525 $\pm$ 0.006

Per-class performance shows the model handles the two extremes of the risk spectrum reasonably well but struggles to isolate the Moderate category:

Class	Precision	Recall	F1-Score	Support
Low	0.57	0.38	0.45	6,261
Moderate	0.22	0.34	0.27	2,453
High	0.65	0.72	0.68	6,143

An ROC-AUC value of 0.718 suggests that the ranking performed by the model is significant (much higher than the random one, which is 0.5), and the model performance proved to be reliable across folds due to the low standard deviation (0.006). Still, a prediction accuracy of about 52% is not a great score and results from two baseline assumptions:

1. The model relies on only 12 constructed features among 25 available in the raw dataset,

2. The three-class risk classification is an artificial one constructed based on the diagnosis and cognitive score and is thus quite ambiguous, especially regarding the Moderate class, which is the most difficult to differentiate from its neighboring classes.

A qualitative check using SHAP showed that the predictions were logically sound: for example, for an individual patient (aged 70, with a positive status of APOE- ϵ 4 and hypertension/depression), the predicted risk category was High (with 45.6%).

Potential Improvements

- Hyperparameter tuning: grid or randomized search over tree depth, number of estimators, and minimum samples per split/leaf to reduce underfitting on the Moderate class.
- Address class definition: refine the Low/Moderate/High labeling rule, or treat risk as a continuous score using predicted probabilities rather than a fixed threshold, to reduce boundary noise between Moderate and its neighbors.
- Expand feature set: incorporate additional variables from the raw dataset (e.g., sleep quality, dietary habits, air pollution exposure, income level) to test whether they add predictive value beyond the 12 Lancet-aligned features currently used.
- Class imbalance handling: apply SMOTE or targeted oversampling to the Moderate class in addition to the balanced class weighting already used.
- Model comparison: benchmark the Random Forest baseline against Gradient Boosting (e.g., XGBoost/LightGBM) and logistic regression to confirm Random Forest remains the best fit before finalizing the architecture.

- Calibration: assess and calibrate predicted probabilities, since the educational risk percentages shown to end users should reflect true confidence levels.

Conclusion

The basic Random Forest model achieves the goal of creating a functional and interpretable risk classification pipeline based on the educational goals of NeuroGuide AI as well as the risk factors uncovered during Week 2 of research (Livingston et al., 2024; Breiman, 2001; Lundberg & Lee, 2017). Despite low values of overall accuracy (0.516) and F1 (0.519), the ROC-AUC value of 0.718 along with stable performance during cross-validation proves that the model is indeed learning and is not just overfitting or guessing. There are several possible improvements outlined below which clearly define the way forward to creating a better model for the following weeks before integration with other parts of the *NeuroGuide AI* system.

AI Disclosure:

Artificial intelligence was used to assist with running and validating the baseline model code, organizing performance metrics, and formatting this report. The model design decisions, feature selection rationale, interpretation of results, and final content were reviewed and verified by the author, who assumes full responsibility for the accuracy of this submission.

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